ALEKSANDR NUZHNYI

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EDUCATION

New York University, New York, NY **May 2027**
Bachelor of Science; Computer Science (Cumulative GPA: 3.89/4.00 | Dean's List)

- Minor: Mathematics
- Relevant coursework: Computer Vision, Software Engineering, Open Source Software Development

Fordham University, New York, NY **June 2023-May 2025**
Bachelor of Science; Computer Science and Mathematics (Cumulative GPA: 4.00/4.00 | Dean's List)

- Minor: Business Administration
- Relevant coursework: Data Structures, Theory of Computation, Business Communications

TECHNICAL SKILLS

Languages: C++, C, JavaScript, Java, Python, HTML, CSS
Frameworks & Libraries: PyTorch, OpenCV, Flask, PyMuPDF, Pytest, NumPy
Tools & Databases: Git, Docker, MongoDB, GitHub Actions (CI/CD)

EXPERIENCE

Web Master, NYU Math Society, New York, NY **June 2026-Present**

- Utilize vanilla **HTML, CSS, and JavaScript** to build and deploy responsive 5-page website on GitHub Pages without initial framework or build step, with a custom domain
- Implement an accessible, **mobile-first UI** including a responsive navigation menu, keyboard/ARIA-labeled carousel controls, lazy-loaded images, and graceful fallbacks when content fails to load
- Engage 70+ members through weekly meetings and manage the society's presence on LinkedIn and NYU channels

Undergraduate Researcher, Regev Lab, New York, NY **March 2026-Present**

- Develop a config-driven **PyTorch training pipeline:** 21-flag CLI, reproducible per-run output dirs, and a subprocess-isolated batch sweep runner
- Build a MAF **parser** that discovers species by genome assembly rather than contig, scaling data ingestion from 1 locus to 6 (58–79 species each)
- Implement the masked-tensor l**oss and partial-checkpoint loader** behind a novel label-free objective for learning interpretable sequence motifs.
- Architect a dependency-free **JS/canvas heatmap over a ~950K-cell grid**, recomputing model activations in-browser with a PyTorch parity check

PROJECT EXPERIENCE

CNN and Image Filtering from Scratch (Python, PyTorch, NumPy) **Fall 2026**

- Built a **CNN without autograd**, writing the backpropagation code by hand for convolution (im2col), max-pooling, linear and cross-entropy layers
- Reached 97.6% test accuracy on MNIST with a 9K-parameter model, and checked the gradients against finite-difference estimates
- Implemented 2D convolution, Gaussian and median filters, and edge detection in plain **NumPy**

FunnAI - Containerized Joke-Scoring Web App (Python, Flask, Docker, GitHub Actions, Gemini API) **Spring 2026**

- Built ML client's humor-analysis module in Python, calling the **Google Gemini API** to classify transcribed speech as a joke and score it 1–100, with input validation and custom exception handling for out-of-range model output
- **Containerized the ML client with a Docker image** (Python 3.10-slim, Pipenv deterministic installs) and wired it into a 3-service Docker Compose stack of Flask web app, ML client, and MongoDB in collaboration with 4 developers
- Configured **GitHub Actions CI** (pytest, pylint, black) to automate the team's workflow

ADDITIONAL INFORMATION

Languages: Russian (Fluent)
Interests: History of Contemporary Fashion, Powerlifting, Hiking, Learning New Subjects